

QUICK FIX: Database Corruption

The Error

SQLITE_CORRUPT: database disk image is malformed

When saving posts with markdown content.

Instant Fix (3 Steps)

Step 1: Stop Backend

Press Ctrl+C in terminal running backend

Step 2: Reset Database

```
cd backend
```

```
# Run reset script  
./reset-database.sh
```

```
# OR manually:
```

```
rm -f local.db local.db-shm local.db-wal  
cargo run --bin setup
```

Step 3: Restart

```
cargo run
```

Done! Database is fresh and working.

This Will Delete All Data

If you need to save your posts first:

Option 1: Export via API (if backend still runs)

```
# While backend is running  
curl http://localhost:3001/api/posts?limit=1000 > posts_backup.json
```

Option 2: Direct Database Export

```
cd backend
```

```
# Try to dump  
sqlite3 local.db ".dump" > backup.sql
```

```
# If that works, restore later:
sqlite3 local_new.db < backup.sql
```

Long-Term Solution: Use Turso

Local SQLite files can corrupt. Turso Cloud is more reliable.

Quick Turso Setup (5 minutes)

```
# 1. Install Turso CLI
curl -sSfL https://get.tur.so/install.sh | bash

# 2. Login
turso auth login

# 3. Create database
turso db create blog-db

# 4. Get credentials
turso db show blog-db --url
turso db tokens create blog-db

# 5. Update backend/.env
# TURSO_DATABASE_URL=libsql://your-db.turso.io
# TURSO_AUTH_TOKEN=your-token-here

# 6. Initialize
cd backend
cargo run --bin setup
cargo run
```

Benefits: - No corruption issues - Cloud backup - Free tier (9GB) - Better performance - No local file issues

Why This Happens

Common Causes: 1. **Improper shutdown** - Backend killed while writing 2. **Disk errors** - Filesystem issues 3. **Concurrent writes** - Multiple updates at once 4. **Large content** - Very long markdown posts

The Fix: - Local SQLite → Single file, can corrupt - Turso Cloud → Distributed, self-healing

Prevention

If Using Local SQLite:

1. Proper Shutdown:

```
# Always use Ctrl+C (SIGINT)
# NOT: kill -9 (force kill)
```

2. Regular Backups:

```
# Add to crontab
0 */6 * * * * sqlite3 /path/to/local.db ".dump" > backup.sql
```

3. File Permissions:

```
# Ensure backend can write
chmod 644 local.db
```

If Using Turso:

Nothing needed! Turso handles everything.

Test After Fix

1. Start Backend

```
cd backend
cargo run
```

2. Test Health

```
curl http://localhost:3001/health
# Should return: {"status":"ok"}
```

3. Create Test Post

1. Login: `http://localhost:3000/login`
2. Create new post
3. Add markdown with special characters:
 - Quotes: `"test"`
 - Apostrophes: `it's`
 - Backticks: ``code``
 - Code blocks
4. Save
5. Should work!

4. Edit Post

1. Go to /admin/posts
2. Click "Edit"
3. Modify content
4. Click "Update Post"
5. Should save without errors

Comparison

Local SQLite	Turso Cloud
Can corrupt	Self-healing
Single file	Distributed
Manual backup	Auto backup
Concurrency issues	Handles concurrency
Simple setup	Simple setup
Free	Free (9GB)

Still Having Issues?

Error Persists After Reset

Try:

```
cd backend
```

```
# Complete clean
```

```
cargo clean
```

```
rm -rf target/
```

```
rm -f local.db*
```

```
# Rebuild
```

```
cargo build
```

```
cargo run --bin setup
```

```
cargo run
```

Turso Connection Fails

Check:

```
# Test connection
```

```
turso db shell blog-db
```

```
# Should open SQL prompt  
# Type: SELECT 1;  
# Should return: 1  
  
# Exit: .exit
```

Posts Still Won't Save

Debug:

```
cd backend  
  
# Run with debug logging  
RUST_LOG=debug cargo run  
  
# Try saving a post  
# Watch console for errors
```

Quick Commands

Reset Everything

```
cd backend  
./reset-database.sh  
cargo run
```

Switch to Turso

```
# See DATABASE_CORRUPTION_FIX.md  
# Section: "Turso Cloud Migration (Detailed)"
```

Backup Current Data

```
cd backend  
sqlite3 local.db ".dump" > backup_$(date +%Y%m%d).sql
```

Restore from Backup

```
cd backend  
rm -f local.db  
sqlite3 local.db < backup_20240201.sql
```

Summary

Immediate Fix: 1. Stop backend 2. Run `./reset-database.sh` 3. Restart backend 4. Working!

Long-term Fix: 1. Set up Turso Cloud 2. Never worry about corruption again 3. Production ready!

Choose your path: - Need it working NOW? → Reset database - Want reliability? → Use Turso - Both? → Reset now, migrate to Turso later

All fixed! Your blog should be working perfectly now.